CUSTOMER SHOPPING BEHAVIOR ANALYSIS

A data preparation and interactive business intelligence project

Project purpose

This project examines customer shopping records to understand purchasing patterns, customer characteristics, product-category activity and review behaviour. It combines Python-based data preparation with a Power BI report designed to make the information easier to explore and communicate.

3,900	18	37	$59.76
Records	Source columns	Missing review ratings before cleaning	Average purchase amount

The dataset contains 3,900 customer purchase records and 18 original fields. The figures above come from the descriptive outputs saved in the supplied notebook. The average purchase amount is reported in USD.

Project at a glance

Tools: Python, Pandas, Jupyter Notebook, SQLAlchemy, PyMySQL, MySQL and Power BI.
Primary deliverables: a data-cleaning and feature-engineering notebook, a Power BI dashboard file, and this stakeholder-facing project report.

1. Business context and objectives

Retail teams need a clear view of who is purchasing, what product groups are being bought, how much customers spend, and how customers rate their shopping experience. This project organises the available customer-level purchase data into a cleaner analytical dataset and presents selected measures through an interactive dashboard.

Project objectives

• Inspect the dataset structure, data types and completeness.
• Resolve missing values in the review-rating field.
• Standardise field names to support consistent analysis and database use.
• Create age segments and a numeric representation of purchase frequency.
• Check whether the discount and promo-code fields contain duplicate information.
• Prepare the transformed data for a MySQL table and Power BI reporting.

2. Dataset overview

The notebook loads a file named **customer_shopping_behavior.csv**. The saved notebook outputs show 3,900 rows and 18 columns. The fields cover customer ID and age, gender, item and category, purchase amount, location, size, colour, season, review rating, subscription status, shipping type, discount and promo-code indicators, previous purchases, payment method and purchase frequency.

Data area	Examples of fields	Analytical use
Customer profile	Age, gender, location, subscription status	Segmenting customers
Purchase details	Item, category, purchase amount, season	Comparing product and spend patterns
Experience & fulfilment	Review rating, shipping type, size, colour	Exploring customer experience and preferences
Purchase behaviour	Previous purchases, payment method, frequency	Understanding repeat-purchase behaviour

Initial profile observations from the notebook: the most frequent category is Clothing (1,737 records); the most frequent gender value is Male (2,652 records); the most frequent subscription-status value is No (2,847 records); and Spring is the most frequent season (999 records). These are category counts, not revenue rankings.

3. Data preparation and methodology

The supplied notebook documents the following preparation workflow:

1. Load and inspect. Pandas is used to load the CSV and inspect sample records, data types, descriptive statistics and missing-value counts.

2. Handle missing review ratings. The initial dataset contains 37 missing values in *Review Rating*. The notebook calculates the median rating within each product category and fills missing ratings using the matching category median. A subsequent null check in the saved notebook output shows zero missing values across the fields.

3. Standardise column names. Column names are converted to lowercase and spaces are replaced with underscores. The purchase amount field is renamed to *purchase_amount_usd*.

4. Create age brackets. The notebook uses quartile-based binning with *pandas.qcut()* to create four groups: Young, Young Adult, Middle-Age and Senior. The saved bin edges are 18, 31, 44, 57 and 70.

5. Convert purchase frequency to days. Frequency labels are mapped to approximate day values, such as Weekly = 7, Fortnightly/Bi-Weekly = 14, Monthly = 30, Quarterly/Every 3 Months = 90 and Annually = 365.

6. Check potentially redundant fields. The notebook compares *discount_applied* and *promo_code_used*. Its equality check returns True for all rows, indicating that the two columns have identical values in this dataset. The notebook then demonstrates dropping *promo_code_used*.

4. Descriptive findings

Measure	Notebook result	Interpretation
Average age	44.07 years	The mean age is in the mid-40s.
Average purchase amount	$59.76	Mean purchase value in USD.
Average review rating	3.75 / 5	Mean rating among the 3,863 non-missing ratings before imputation.
Previous purchases	25.35 average	Mean number of previous purchases recorded per customer record.
Age range	18–70 years	Observed minimum and maximum ages.

These descriptive statistics are taken from the saved notebook outputs. The mean review rating shown above is the pre-imputation value (3.7501 across 3,863 available ratings). No claim is made here about a post-cleaning average rating.

5. Power BI dashboard

The supplied Power BI file is named **Retail_business_dashboard.pbix**. Its report layout references a *customer_data* table and includes summary cards and category/age-related visuals for exploring customer purchase behaviour.

Dashboard elements documented in the supplied report

• KPI cards: Total Customers, Average Purchase Amount and Average Review Rating.
• Category analysis: purchase amount/revenue by category and customer/sales volume by category.
• Age analysis: revenue and sales by age group.
• Subscription analysis: customer share by subscription status.
• Slicers: subscription status, gender, category and shipping type.

The slicers allow a viewer to narrow the dashboard to selected customer groups and fulfilment or product dimensions. This supports comparisons without changing the underlying dataset manually.

6. Data flow and technical implementation

The intended workflow links preparation, storage and reporting:

Customer shopping CSV 18 original fields	**Python + Pandas** Inspection, cleaning, derived fields	**MySQL** retail_dataset.customer_data	**Power BI** Interactive dashboard

The notebook contains code to connect to a local MySQL database using SQLAlchemy and PyMySQL and to write the transformed DataFrame to a table named *customer_data* in *retail_dataset*. The saved notebook does not provide evidence that this database upload completed successfully. The SQL upload cell also contains an argument typo (*ifexists* rather than *if_exists*) that should be corrected before execution. The connection credentials shown in the notebook should be replaced with secure local credentials and never committed to a public repository.

7. Key takeaways

• The notebook starts with a structured dataset of 3,900 customer purchase records.
• Review Rating is the only field with missing values in the initial null check; 37 values are imputed using category-level medians.
• Age brackets and approximate purchase-frequency days are engineered to make segmentation and behavioural comparisons easier.
• Discount Applied and Promo Code Used are identical across the supplied records, so one may be redundant for this dataset.
• The Power BI report is structured around customer, category, age and subscription views, with interactive slicers.

8. Limitations and next steps

The project files supplied for this report include the notebook, the Power BI report and a separate Excel workbook. The Excel workbook contains a *Sales_data* sheet with a different sales/order-oriented schema; it does not match the customer-shopping dataset referenced by the notebook and dashboard, so it has not been treated as an input to this project.

The original CSV file referenced by the notebook was not included among the supplied files. As a result, the notebook's saved outputs can be used for the descriptive facts in this report, but the complete analysis could not be independently rerun from the provided files. The Power BI file's layout and field references can be inspected, but refreshed dashboard values were not available for independent validation. For this reason, this report does not invent dashboard KPI totals or claim category revenue leaders.

Recommended improvements

1. Include the source CSV (where licensing permits) or document a stable download source and update the notebook path.
2. Correct and test the MySQL upload cell, use environment variables for credentials, and record a successful row-count check after loading.
3. Validate the assumption that identical discount and promo-code columns are truly interchangeable before removing one from a production dataset.
4. Document the definitions of dashboard measures such as Total Customers, Sales and Revenue, including whether customer IDs are counted distinctly or as rows.
5. Add a date dimension or time-based fields if transaction dates are available, enabling trend and seasonality analysis.
6. Add a concise dashboard insight section after refreshing the report with the correct source data and confirming its displayed KPI values.

9. Conclusion

This project demonstrates a practical analytics workflow from raw customer-shopping data to a prepared analytical table and an interactive Power BI report. The Python notebook covers data inspection, missing-value treatment, column standardisation and feature engineering. The dashboard provides a visual interface for exploring customer and purchasing patterns by category, age group and subscription status. The next step is to validate the database load and refresh the dashboard against the documented source dataset so that stakeholders can rely on the final KPI values.

Prepared from the supplied project files: *Untitled1.ipynb* and *Retail_business_dashboard.pbix*. Descriptive values are quoted from saved notebook outputs. Where source data or refreshed dashboard values were unavailable, the limitation is stated rather than estimated.